Where is the Land of Opportunity Replication

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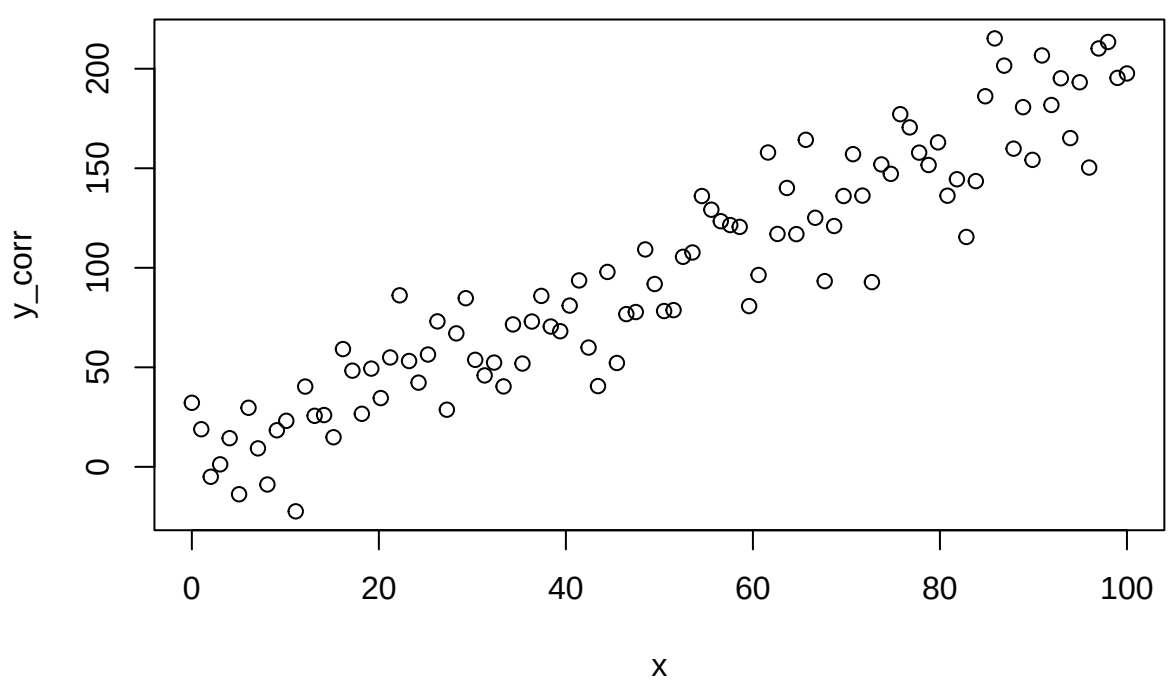
Testing out the libraries

Small demonstration of the package functionality.

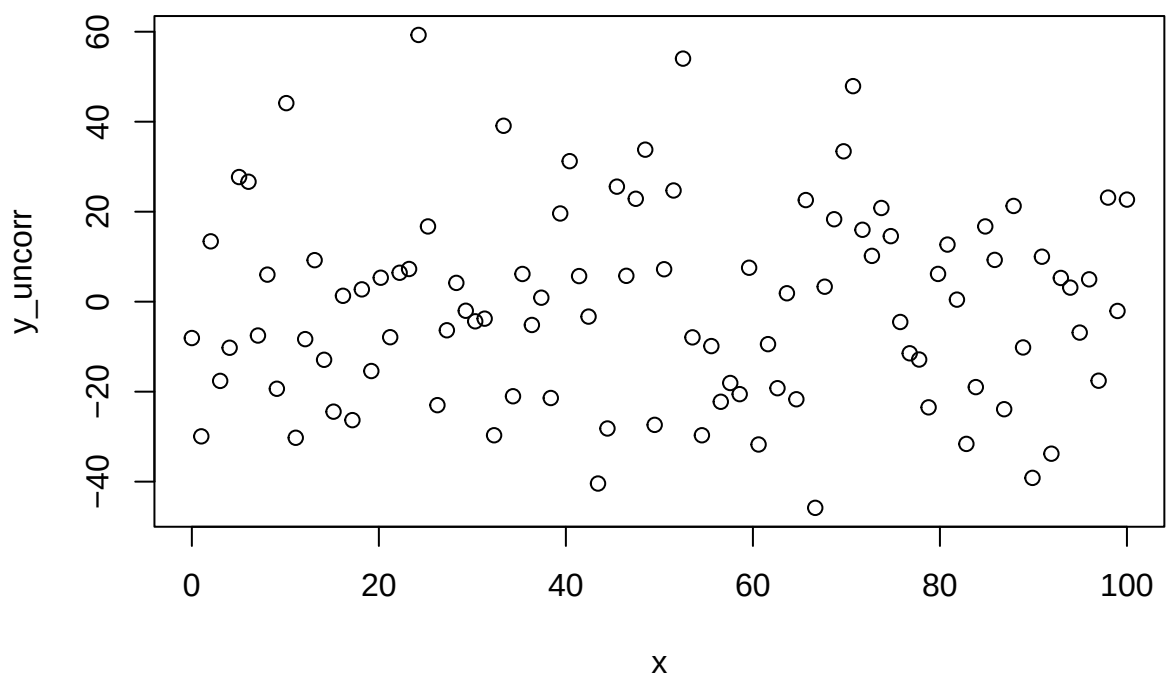
```
# Uncomment to install Perk (package that implements Diciccio and Romano paper)
# library("devtools")
# install_github("hyu-ub/perk/perk")
library(perk)
# Obtained from https://github.com/hyu-ub/perk

# Correlated data
x <- seq(from = 0, to = 100, length.out = 100)
y_corr <- 2 * x + 1 + rnorm(100, sd = 20)
plot(x, y_corr)
```



```r
perk_test(x, y_corr, method = "pearson", alternative = "two.sided")
```

```
## $estimate
## [1] 0.9417102
##
## $p.value
## [1] 0
##
## $method
## [1] "pearson"
##
## $alternative
## [1] "two.sided"
```

```r
# Uncorrelated data
y_uncorr <- rnorm(100, sd = 20)
plot(x, y_uncorr)
```




```r
perk_test(x, y_uncorr, method = "pearson", alternative = "two.sided")
```

```
## $estimate
## [1] 0.001776648
##
## $p.value
## [1] 0.99
##
## $method
```

```
## [1] "pearson"
##
## $alternative
## [1] "two.sided"
```

The following package also implements the test:

https://cran.r-project.org/web/packages/PermCor/PermCor.pdf

It has basically the same interface.

```r
# Uncomment to install PermCor (package that implements Diciccio and Romano paper)
# install.packages("PermCor")
library(PermCor)

perm_test(x, y_corr, method = "Pearson", alternative = "two.sided")
```

```
## $estimate
## [1] 0.9417102
##
## $p.value
## [1] 0
##
## $method
## [1] "Pearson"
##
## $alternative
## [1] "two.sided"
```

```r
perm_test(x, y_uncorr, method = "Pearson", alternative = "two.sided")
```

```
## $estimate
## [1] 0.001776648
##
## $p.value
## [1] 0.99
##
## $method
## [1] "Pearson"
##
## $alternative
## [1] "two.sided"
```

The results are the same for the correlated data, and a bit different (not much) for the uncorrelated data. We can try both and if results differ significantly, investigate further.

Applications in linear regression

In this section, I explore how to use the permutation tests to test the significance of a linear regression. I wanted to understand how to do this since it seems we will likely have to code this ourselves.

Consider a model with the following assumptions:

(1) $Y_i = \alpha + X_i^T \beta + \epsilon_i, i = 1, ..., n$ where $X_i \in R^p$ is a vector of predictor variables, and ϵ_i is a mean zero error term.

(2) (Y_i, X_i) are i.i.d. according to a distribution P with $E(\epsilon_i \cdot X_i) = 0$.

(3) $\Sigma_{XX} := E((X_i - \mu X))$ and $\Omega := E(\epsilon_i^2 (X_i - \mu X)(X_i - \mu X)^T)$ are nonsingular. Furthermore, $\Sigma_{i=1}^n X_i X_i^T$ is almost surely invertible.

According to the paper, we can use this to test $H_0 : \beta = 0$ (there is no relation between the variables).

The test statistic is: $S_n(X, Y) = n\hat{\beta}_n^T \left(\hat{\Sigma}_{XX}^{-1} \hat{\Omega} \hat{\Sigma}_{XX}^{-1} \right) \hat{\beta}_n$ where $\hat{\Sigma}_{XX} = \frac{1}{n} \Sigma_i X_i^T X_i$ and $\hat{\Omega} = \frac{1}{n} \Sigma_i Y_i^2 X_i X_i^T$.

We also assume some things such as the X_is and Y_is having a sample mean of 0 (we can normalize) and that our regression model has a constant term (I believe they mean the intercept).

Theorems 3.2 and 3.3 describe ways to actually perform the test under different assumptions.

Theorem 3.2: Test 1- Permuting the Y_is

Assumes (1)-(3), $\beta = 0, E(Y_1^4) < \infty, E(X_{1j}^4) < \infty, j = 1, ..., p$.

An advantage of this test is that we can extend the test to the situation where only a subset of the regression coefficients are 0. This is covered in 3.4 and later.

Theorem 3.2 with extra assumptions

If we additionally assume that the e_is are symmetric (in a group-theoretic sense or around the mean or what?) and independent of the X_is, then we can obtain an exact permutation test.

"This test will not be exact if there is dependence, but since the statistic is asymptotically pivotal, we expect that the permutation test will be asymptotically level α." (I don't know what asymptotically pivotal is, but the rest is pretty clear).

Theorem 3.3: Test 2- Changing the sign of the Y_is

Assumes (1)-(3), $\beta = 0$.

Here we can flip the sign of the Y_is and permute them to create an exact level test.

Comparison and Discussion

When $\beta = h/\sqrt{n}$, the permutation distribution of S_n of both tests still behaves well asymptotically under both methods. On the other hand, the sampling distribution of S_n, does not behave that well.

"... the two tests have the same limiting local power functions under such alternative sequences." To the best of my understanding, they perform the same, but 3.2 can be extended to more scenarios, so it would be beneficial to use that one. The extra assumptions of 3.2 (when compared to 3.3) do not seem like they would matter in any real world application.

Testing if only a subset of the regression coefficients are 0

TODO: Need to work on this. Do after fully reading the paper as it may not be necessary.

Tangent on Permutation vs Bootstrap

"Permutation-based analyses resemble the bootstrap in that they rely on randomizations of the observed data. The primary difference is that while bootstrap analyses typically seek to quantify the sampling distribution of some statistic computed from the data, permutation analyses typically seek to quantify the null distribution." (https://pillowlab.princeton.edu/teaching/mathtools16/slides/lec21_Bootstrap.pdf)

The permutation has fewer assumptions than a regular hypothesis test because instead of assuming a theoretical distribution (like the t-test or any other parametric test), it builds its own permutation distribution. The only major assumption is that of exchangeability, which can sometimes be circumvented if the test statistic is properly studentized (that's the whole Diciccio and Romano paper).

Correlation permutation function

```r
# Permutation function based on theorem 3.2.
# Tests whether or not a regression is significant. The null is that beta = 0.
# The Xs and Ys MUST be standardized to have sample mean zero.
# X: A matrix of predictor variables.
# Y: A vector of the response variable.
# B: Number of permutations. Default of 10000
# seed: For reproducibility.

# Supports multiple predictor variables.
# It always fits the intercept based on remark 3.1. Otherwise, our results could be unreliable.
perm_test_regression <- function(X, Y, B = 10000, seed = NULL) {
  if (!is.null(seed)) {
    set.seed(seed)
  }

  # Ensure matrix form
  X <- as.matrix(X)
  Y <- as.numeric(Y)
  n <- nrow(X)
  if (n != length(Y)) {
    stop("X and Y must have same number of rows")
  }

  # Always include an intercept
  X <- cbind(1, X)
  p <- ncol(X)

  # Function to compute the test statistic S_n
  compute_Sn <- function(X, Y) {
    XtX <- crossprod(X)           # t(X) %*% X  (p x p)
    XtY <- crossprod(X, Y)        # t(X) %*% Y  (p x 1)
    beta_hat <- solve(XtX, XtY)   # OLS estimate (p x 1)
    resid <- as.vector(Y - X %*% beta_hat)
    Sigma_hat <- XtX / n
    r2 <- resid^2
    # Omega^ = (1/n) * X' diag(r2) X
    Omega_hat <- crossprod(X, X * r2) / n
    tmp <- solve(Sigma_hat, beta_hat)
    as.numeric(n * t(tmp) %*% Omega_hat %*% tmp)
  }

  # Observed statistic
  S_obs <- compute_Sn(X, Y)

  # Permutation distribution
  S_perm <- replicate(B, {
    Y_perm <- sample(Y, replace = FALSE)
    compute_Sn(X, Y_perm)
  })

  # Right-tailed p-value with +1 correction
  pval <- (sum(S_perm >= S_obs) + 1) / (B + 1)
```

```
  list(
    S_obs = S_obs,
    S_perm = S_perm,
    p_value = pval,
    B = B
  )
}
```

Test correlation permutation function

```
# Turn the xs and ys into mean 0
x_m0 <- x - mean(x)
y_corr_m0 <- y_corr - mean(y_corr)
y_uncorr_m0 <- y_uncorr - mean(y_uncorr)

# Test the correlated variables
perm_results <- perm_test_regression(x_m0, y_corr_m0)
perm_results$p_value
```

```
## [1] 0.00089991
```

```
# Test the uncorrelated variables
perm_results <- perm_test_regression(x_m0, y_uncorr_m0)
perm_results$p_value
```

```
## [1] 0.9875012
```

We obtain great p-values!

Paper Notes

Section 1: Introduction

Studies the intergenerational mobility in the US. This is mostly done by looking at kid's wealth compared to parent's wealth.

Analysis is mostly done across "Commute Zones" which is like metropolitan zones that also account for rural areas.

In general, a lot of variance is found. In the best areas, as good as Canada and Denmark while lower levels are lower than any developed country in the dataset. A lot of interesting relationships of mobility with other variables are explored. This is especially true when mobility is decomposed into relative mobility and absolute mobility.

Section 2: Measures of Intergenerational Mobility

Relative Mobility:

"What are the outcomes of children from low-income families relative to those of children from high-income families?"

Two measures are used:

(1) Intergenerational Income Elasticity (IGE): $\left(\frac{dE[\log(Y_i)|X_i=x]}{d\log(x)}\right)$

The way to estimate the IGE is to regress log child income ($\log(Y_i)$) on log parent income ($\log(X_i)$), yielding a coefficient of:

$$IGE = \rho_{XY} \frac{SD(\log(Y_y))}{SD(\log(X_i))}$$

where $\rho_{XY} = Corr(\log(X_i), \log(Y_i))$. (yes, Pearson)

(2) Rank-Rank Slope: Another way to quantify relative mobility is to measure the (Pearson) correlation between child and parent ranks. $\rho_{PR} = Corr(P_i, R_i)$.

These two metrics are scale-invariant measures of how much child income depends on parent income. For some differences in interpretation, look at the paper. The authors use both measures but focus primarily focus on the rank-rank slope due to it being "much more robust".

Absolute Mobility:

"What are the outcomes of children from families of a given income level in absolute terms?"

Three measures are used:

(1) Absolute Upward Mobility: "the mean rank (in the national child income distribution) of children whose parents are at the 25th percentile of the national parent income distribution."

At the national level, this is a slightly downgraded rank-rank slope. At the smaller, local level, this is more useful.

(2) The probability of rising from the bottom quintile to the top quintile in income.

(3) The probability that a child has family income above the poverty line conditional on having parents at the 25th percentile.

Discussion:

Based on my reading so far, it seems we will likely be testing the relative mobility more than the absolute mobility.

Section 3: Data

The data used is US citizen federal tax records spanning 1996-2012. So we are mostly looking at children born between 1980-1991.

There are 10M kids in the "core sample" and 44M kids in the "extended sample". Definitions are found in the paper.

The earnings of children can only be reliably measured up to age 32 (sample started with 1980 cohort and study came out in the early 2010s).

Definitions are carefully reviewed in the rest of the section.

Section 4: National Statistics

Starting here, tables and figures are cited. There seem to be multiple Figure 1s and 2s and so forth. They are different, and I'm not 100% sure which one I'm supposed to look at (though I believe the ones that come first). The same happens with the tables.

Ok so the second one is Online Appendix C I believe?

Baseline Estimates

Core sample is used here.

Figure 1A is a bit tricky to understand at first, but if they based regressions on this, it would seem a mistake (non-linear relationship). They even admit this in the third paragraph of page 13, and then they run the log-log regression. CHECK THIS.

The IGE changes dramatically on what percentile cutoffs are used and how children with $0 income are handled. We could maybe do something here?

We could likely do a fair bit of significance testing for Table 1. CHECK THIS.

For the rank-rank estimates, we could also arguably do some replication. It might not make the most sense since the relationship between the ranks is nearly exactly linear, so testing the significance with different methods might not be that relevant. CHECK THIS.

Some other interesting work is done with transition matrices, but I doubt we can apply permutation methods here.

Robustness of Baseline Estimates

We could arguably test for significance each point of figure 3a. Same thing with Online Appendix Figure IIb (though not as interesting?). Honestly, a lot of the stuff here we could replicate. CHECK THIS.

They don't talk much about statistical robustness.

Intermediate Outcomes: College Attendance and Teenage Birth

There are more regressions here for which we could test significance.

Section 5: Spatial Variation in Mobility